VERSALITY OF X AT U_0 MATCHES VERSALITY OF THE FORMAL OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1902-1914 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 1916-1949, the complete author proof (32 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 33 of queue_artin.json (pb=1251, stmt 1902-1914, proof 1916-1949).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 07XF: sources/stacks-artin/pinned-original-source.tex, source lines 1887–1896. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07X6: sources/stacks-artin/pinned-original-source.tex, source lines 1289–1295. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06GW: sources/stacks-artin/pinned-original-source.tex, source lines 1949–1949. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06HC: sources/stacks-artin/pinned-original-source.tex, source lines 1949–1949. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:artin.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.1. *With notation as in Definition 07XF. Let $R = \mathcal{O}_{U,u_0}^\wedge$. Let ξ be the formal object of $\mathcal{X}$ over R associated to $x|_{\mathrm{Spec}(R)}$, see (07X6). Then*

$$x \text{ is versal at } u_0 \Leftrightarrow \xi \text{ is versal}$$

Proof. Observe that $\mathcal{O}_{U,u_0}$ is a Noetherian local S-algebra with residue field k. Hence $R = \mathcal{O}_{U,u_0}^\wedge$ is an object of $\mathcal{C}_\Lambda^\wedge$, see Formal Deformation Theory, Definition 06GW. Recall that ξ is versal if $\underline{\xi} : \underline{R}|_{\mathcal{C}_\Lambda} \to \mathcal{F}_{\mathcal{X},k,x_0}$ is smooth and x is versal at u_0 if $\hat{x} : \mathcal{F}_{(Sch/U)_{fppf},k,u_0} \to \mathcal{F}_{\mathcal{X},k,x_0}$ is smooth. There is an identification of predeformation categories

$$\underline{R}|_{\mathcal{C}_\Lambda} = \mathcal{F}_{(Sch/U)_{fppf},k,u_0},$$

see Formal Deformation Theory, Remark 06HC for notation. Namely, given an Artinian local S-algebra A with residue field identified with k we have

$$\mathrm{Mor}_{\mathcal{C}_\Lambda^\wedge}(R, A) = \{\varphi \in \mathrm{Mor}_S(\mathrm{Spec}(A), U) \mid \varphi|_{\mathrm{Spec}(k)} = u_0\}$$

Unwinding the definitions the reader verifies that the resulting map

$$\underline{R}|_{\mathcal{C}_\Lambda} = \mathcal{F}_{(Sch/U)_{fppf},k,u_0} \xrightarrow{\hat{x}} \mathcal{F}_{\mathcal{X},k,x_0},$$

is equal to $\underline{\xi}$ and we see that the lemma is true. □